

1) Map ::

Home (Ghah Faisal Colony)
↓ 2.5 km

R1 (Dugh road)

↓ 4.5 km

R2 (Shahrah-e-Faisal)

6 km ↙ 5 km ↘

R3

(Rashid Rd Minhas)

R5 (Johar Moh)

↓ 3.5

↓ 4 km

R4 (Uni road)

R4 (Uni road) ←

↓ 1.5 km

Fuvast

Connections ::

Home → R1

R1 → R2

R2 → R3, R5

R3 → R4

R5 → R4

R4 → Fuvast

2) Heuristic values $h(n)$::

Node	$h(n)$
Home	17.0
R1	14.5
R2	10.0
R3	5.5
R5	5.0
R4	1.5
FUVAST	0

Cost Assumption:-

Each road Step = 1

Search :-

Step 0: Initialize

PQ : [Home]

$V = \emptyset$

$g = 0$

$f = 5.2$

$\because g+h = 0 + 5.2 = 5.2$

Parent Table :-

Home $\rightarrow$ None

Step 1: Expand Home.

Neighbours : R1

PQ : [R1]

; $V = [\text{Home}]$

Parent Table Update:

Node	g	h	f
R1	1	5.2	6.2

$$f(R1) = g + h = 1 + 5.2 = 6.2$$

Choose R1 ($f = 6.2$).

Parent Table Update:

Home $\rightarrow$ None

R1 $\rightarrow$ Home

Step 02: Expand R1.

Neighbour : R2

PQ : [R2]

V : [Home, R1]

Node	g	h	f
R2	2	4.0	6.0

$$f(R2) = 2 + 4.0 = 6.0$$

Choose R2 ($f = 6.0$)

Parent Table Update:

Home $\rightarrow$ None

R1 $\rightarrow$ Home

R2 $\rightarrow$ R1

Step 03: Expand R2

Neighbours: R3, R5

PQ = [R3, R5]

V = [Home, R1, R2]

For R3:

$g = 3$, $h = 3.3$

$f = 3 + 3.3 = 6.3$

For R5:

$g = 3$, $h = 4.8$

$f = 3 + 4.8 = 7.8$

Node	g	h	f
R3	3	3.3	6.3
R5	3	4.8	7.8

Choose R3 ($f = 6.3$)

• Parent Table Update:

Home $\rightarrow$ Home

R1 $\rightarrow$ Home

R2 $\rightarrow$ R1

R3 $\rightarrow$ R2

R5 $\rightarrow$ R2

Step 04: Expand R3

Neighbours: R4

$PQ = [R_4, R_5]$

$V: [Home, R_1, R_2, R_3]$

• Parent Table Update: for R_4 :

Home $\rightarrow$ none $g=4, h=1.0$

$R_1 \rightarrow$ Home $f=4+1.0=5.0$

$R_2 \rightarrow R_1$ For R_5 :

$R_3 \rightarrow R_2$ $g=3, h=4.8$

$R_5 \rightarrow R_2$ $f=3+4.8=7.8$

$R_4 \rightarrow R_3$

choose R_5 ($f=5.0$)

Step 05: Expand R_4

Neighbours: FUVAST

$PQ: [FUVAST, R_5]$

$V: [Home, R_1, R_2, R_3, R_4]$

For Fuvast:

$g=5, h=0$

$f=5+0=5.0$

For R_5 :

$g=3, h=4.8$

$f=3+4.8=7.8$

Node	g	h	f
FUVAST	5	0	5.0
R_5	3	4.8	7.8

choose Fuvast. (goal found!)
($f=5.0$)

Step 06: Goal found.

Final Parent Table.

Node	Parent
Home	None
R1	Home
R2	R1
R3	R2
R5	R2
R4	R3
Forest	R4

Start from goal & go backward.

Forest $\rightarrow$ R4 $\rightarrow$ R3 $\rightarrow$ R2 $\rightarrow$ R1 $\rightarrow$ Home

Reverse it:

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ Forest

Total Cost = 17 Km